

INST414 Week 10 Answers

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Probability of a New Movie Fitting the Cluster

Sci-Fi Movie

```
... Pr[Cluster 00 | Sci-Fi]:      0.1734597156398104
    Pr[Cluster 01 | Sci-Fi]:      0.040758293838862564
    Pr[Cluster 02 | Sci-Fi]:      0.20379146919431282
    Pr[Cluster 03 | Sci-Fi]:      0.024644549763033177
    Pr[Cluster 04 | Sci-Fi]:      0.05308056872037915
    Pr[Cluster 05 | Sci-Fi]:      0.0
    Pr[Cluster 06 | Sci-Fi]:      0.06161137440758295
    Pr[Cluster 07 | Sci-Fi]:      0.0
    Pr[Cluster 08 | Sci-Fi]:      0.008530805687203791
    Pr[Cluster 09 | Sci-Fi]:      0.021800947867298578
    Pr[Cluster 10 | Sci-Fi]:      0.0009478672985781991
    Pr[Cluster 11 | Sci-Fi]:      0.047393364928909956
    Pr[Cluster 12 | Sci-Fi]:      0.18862559241706164
    Pr[Cluster 13 | Sci-Fi]:      0.013270142180094787
    Pr[Cluster 14 | Sci-Fi]:      0.03033175355450237
    Pr[Cluster 15 | Sci-Fi]:      0.1317535545023697
```

Fantasy Movie

```
... Pr[Cluster 00 | Fantasy]:      0.23831242873432154
    Pr[Cluster 01 | Fantasy]:      0.009122006841505131
    Pr[Cluster 02 | Fantasy]:      0.18814139110604333
    Pr[Cluster 03 | Fantasy]:      0.033067274800456105
    Pr[Cluster 04 | Fantasy]:      0.05473204104903078
    Pr[Cluster 05 | Fantasy]:      0.002280501710376283
    Pr[Cluster 06 | Fantasy]:      0.16305587229190424
    Pr[Cluster 07 | Fantasy]:      0.0
    Pr[Cluster 08 | Fantasy]:      0.039908779931584946
    Pr[Cluster 09 | Fantasy]:      0.02280501710376283
    Pr[Cluster 10 | Fantasy]:      0.002280501710376283
    Pr[Cluster 11 | Fantasy]:      0.01710376282782212
    Pr[Cluster 12 | Fantasy]:      0.0992018244013683
    Pr[Cluster 13 | Fantasy]:      0.07069555302166476
    Pr[Cluster 14 | Fantasy]:      0.0387685290763968
    Pr[Cluster 15 | Fantasy]:      0.020524515393386546
```

Adventure Movie

```
... Pr[Cluster 00 | Adventure]: 0.28434313137372524
    Pr[Cluster 01 | Adventure]: 0.007198560287942413
    Pr[Cluster 02 | Adventure]: 0.3923215356928614
    Pr[Cluster 03 | Adventure]: 0.08098380323935213
    Pr[Cluster 04 | Adventure]: 0.011397720455908816
    Pr[Cluster 05 | Adventure]: 0.01259748050389922
    Pr[Cluster 06 | Adventure]: 0.07438512297540492
    Pr[Cluster 07 | Adventure]: 0.0
    Pr[Cluster 08 | Adventure]: 0.00599880023995201
    Pr[Cluster 09 | Adventure]: 0.0011997600479904016
    Pr[Cluster 10 | Adventure]: 0.008998200359928014
    Pr[Cluster 11 | Adventure]: 0.004199160167966407
    Pr[Cluster 12 | Adventure]: 0.014397120575884822
    Pr[Cluster 13 | Adventure]: 0.047390521895620874
    Pr[Cluster 14 | Adventure]: 0.010797840431913617
    Pr[Cluster 15 | Adventure]: 0.04379124175164967
```